

计算系统生物学作业1

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说明：由于某些英文表达本人实在不懂，但是作业由英文给出，因此在本人的作业中将以英文为主，辅以中文解释（有可能是那个英文表达我不会）。

1. a.



Transcription转录: The gene (G) is transcribed into mRNA (M) at a production rate of β_g . The gene remains unchanged in the process.



mRNA Degradation mRNA自行分解: The mRNA (M) is degraded and removed from the system at a rate of α_m .



Translation翻译: The mRNA (M) is translated by 核糖体ribosomes into the protein (X) at a production rate of β_m . The mRNA remains unchanged during this translation event.



Ligand Binding配体结合: The protein (X) reversibly binds to the ligand (S) to form a protein-signal complex (X_S). k_{1+} is the forward binding rate, and k_{1-} is the unbinding (dissociation) rate.



Activation Change激活反应: The bound protein complex (X_S) undergoes a reversible structural or chemical change to become the active form (X_S^*). k_{2+} and k_{2-} represent the forward and reverse reaction rates of this activation step.即 X_S 经历的活化过程。

b.

Gene (G): Since the gene is neither created nor destroyed in these reactions, its concentration remains constant. G 只参与了转录，无生成或者消耗。

$$\frac{d[G]}{dt} = 0 \quad (6)$$

mRNA (M): mRNA is produced by transcription and removed by degradation.涉及mRNA的反应有转录（生成），自分解（消耗）和翻译（不改变浓度）

$$\frac{d[M]}{dt} = \beta_g[G] - \alpha_m[M] \quad (7)$$

Protein (X): Protein X is produced by translation of M . It is consumed when it binds to S and is released when the complex X_S dissociates. 涉及到蛋白质 X 的反应有翻译（生成），配体结合（同时有生成和消耗）。这里认为活化不改变浓度，且不存在蛋白质自分解（题目中没有说）。

$$\frac{d[X]}{dt} = \beta_m[M] - k_{1+}[X][S] + k_{1-}[X_S] \quad (8)$$

c.

Reaction	Transcription	mRNA Degradation	Translation	Ligand Binding	Activation Change
Bacteria	~1min	2-5min	~2min	~1ms	1-100 μ s
Mammals	~30min	~10min to over 10h	~30min	~1s	1-100 μ s

d.

由题意转录需要活化蛋白与promoter相结合之后才可以进行

Promoter Binding:



Activated Transcription:



2. a. 注意到这里细胞的体积呈指数增长。且这里每隔 T_D 时间，体积翻倍(doubling time)。

$$V(t) = V_0 e^{\frac{t}{T_D} \ln 2} \quad (11)$$

从而

$$\gamma = \frac{\ln 2}{T_D} \quad (12)$$

b. According to the derivative formula, we have

$$\frac{d[X]}{dt} = \frac{d}{dt} \left(\frac{n}{V} \right) = \frac{V \frac{dn}{dt} - n \frac{dV}{dt}}{V^2} = \frac{1}{V} \cdot \frac{dn}{dt} - \frac{n}{V} \cdot \gamma \quad (13)$$

(Notice that $\frac{dV}{dt} = V_0 \gamma e^{\gamma t}$)

hence

$$\frac{d[X]}{dt} = \frac{k(t)}{V} - \gamma \cdot [X] \quad (14)$$

The decay term comes from cell growth.衰减项源于细胞体积增大

c. Let's take the degradation of the protein into account. (考虑蛋白质自行分解) We modify our equation to

$$\frac{d[X]}{dt} = \frac{k(t)}{V} - \gamma[X] - \delta[X] \quad (15)$$

Set $\frac{d[X]}{dt} = 0$ to reach the steady state.

$$[X]_{ss} = \frac{k(t)}{V(\gamma + \delta)} \quad (16)$$

where $V = V_0 e^{\gamma t}$ and $\gamma = \frac{\ln 2}{T_D}$

d. Here we focus on E.coli. $T_D = 20\text{min}$, $t_{0.5} = 2\text{h}$, $V_0 = 1\mu\text{m}^3$. 色氨酸加氧酶产生周期2min. 直接带入可得

$$[X]_{ss} = \frac{k(t)}{V(\gamma + \delta)} = 12.4\mu\text{M} \quad (17)$$

Reference:生物化学简明教程 (张丽萍、杨建雄)

3. a. According to the image, we have

$$K_1 = \frac{[D][X]}{[DX_A]}, K_2 = \frac{[DX_A][X]}{[DX_A X_B]}, K_3 = \frac{[D][X]}{[DX_B]}, K_4 = \frac{[DX_B][X]}{[DX_A X_B]} \quad (18)$$

hence

$$\frac{[DX_AX_B]}{[D][X]^2} = \frac{1}{K_1K_2} = \frac{1}{K_3K_4} \quad (19)$$

Notice that $K_1K_2 = K_3K_4$, the four constants cannot be independently specified.

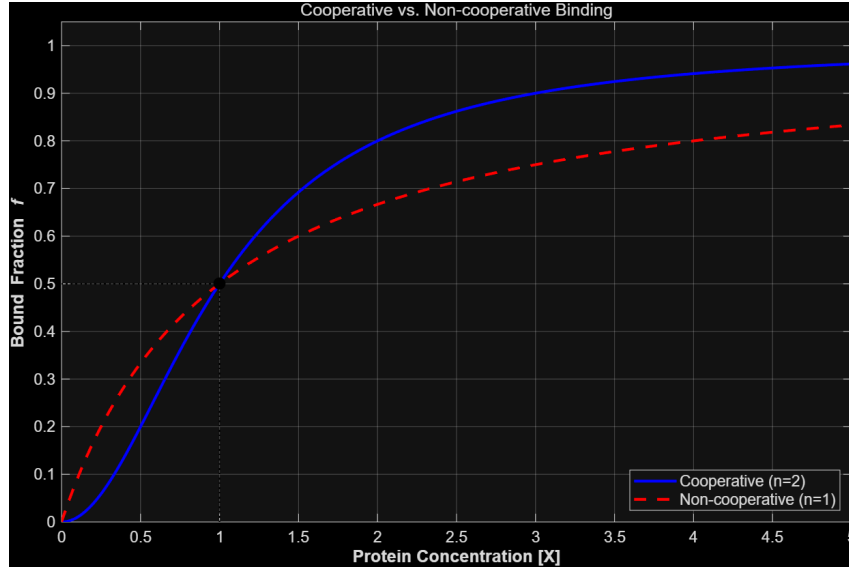
b. 由 $K_1 \gg K_2$, 我们有 $[DX_A]^2 \ll [D][DX_AX_B] \leq (\frac{[D]+[DX_AX_B]}{2})^2 \leq \frac{[D_{tot}]^2}{4}$, $\frac{[DX_A]}{[D_{tot}]} \ll \frac{1}{2}$. Similarly we have $\frac{[DX_B]}{[D_{tot}]} \ll \frac{1}{2}$. Hence $[DX_A]$ and $[DX_B]$ are negligible.

c.

$$f = \frac{[DX_AX_B]}{D_{tot}} = \frac{[X]^2}{[X]^2 + (K_2 + K_4)[X] + K_1K_2} = \frac{\frac{[X]^2}{K_1}}{\frac{[X]^2}{K_1} + \frac{K_2+K_4}{K_1}[X] + K_2} \quad (20)$$

Notice that $K_1 \gg K_2, K_3 \gg K_4$, we have $K_1 = K_4 \frac{K_1}{K_4} \gg K_3 \frac{K_4}{K_1} = K_2 \implies K_1 \gg K_4$

$$f \approx \frac{[X]^2}{[X]^2 + K_1K_2} \quad (21)$$



有一个阈值：阈值以下反应没那么充分，但高于阈值时就会结合的非常充分

4.

4.1 a. Set

$$\frac{d[ES]}{dt} = k_{1+}[E][S] - k_{1-}[ES] = 0 \quad (22)$$

i.e.

$$[ES] = 10[E][S] \quad (23)$$

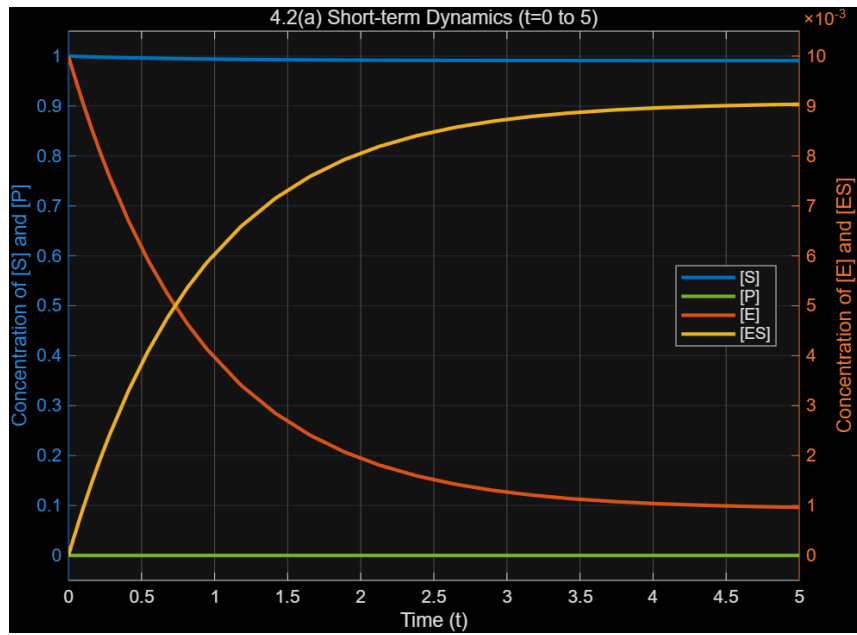
let $[ES] = x$, then we have $10x^2 - 11.1x + 0.1 = 0$, $x = 0.0091$

b. We notice that $\frac{d[ES]}{dt} = [ES]^2 - 1.11[ES] + 0.01$. Therefore, we have $[ES] = -\frac{1.0918}{120.9780e^{1.0918t} - 1} + 0.0091$. Let $-\frac{1.0918}{120.9780e^{1.0918t} - 1} + 0.0091 < 0.0001$, we have $t > 4.1241$. Hence we consider $t = 4.1241$ as the time the reaction reaches quasi-equilibrium.

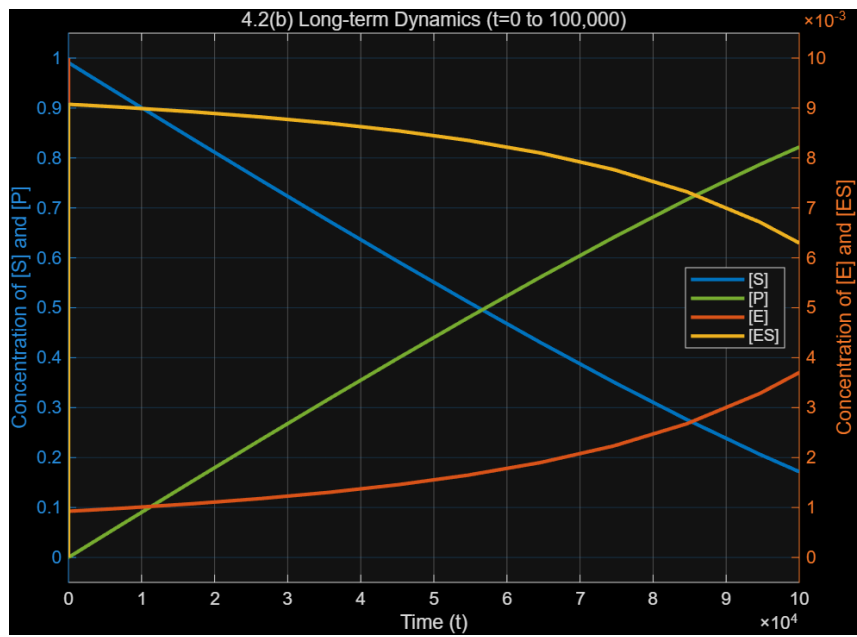
c. We need to analyze the situation where $[S]=0.5$. This implies $[P] \approx 0.5$, since $[E] + [ES] = E_0 = 0.01 \ll 0.5$

Notice that $\frac{d[P]}{dt} = k_2[ES] \approx 0.000001$, we have $t = 50000$

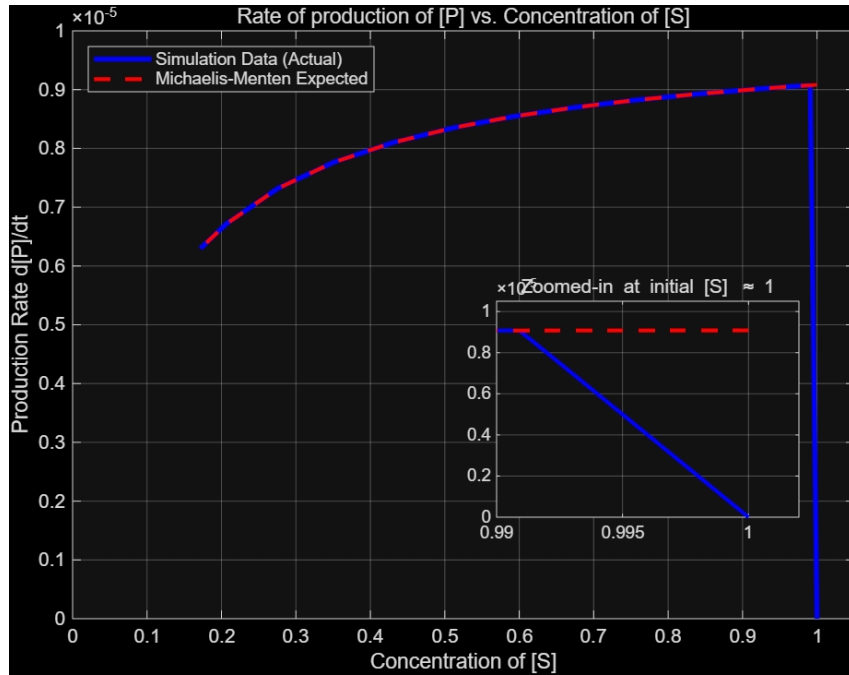
4.2 a.



b.



c.

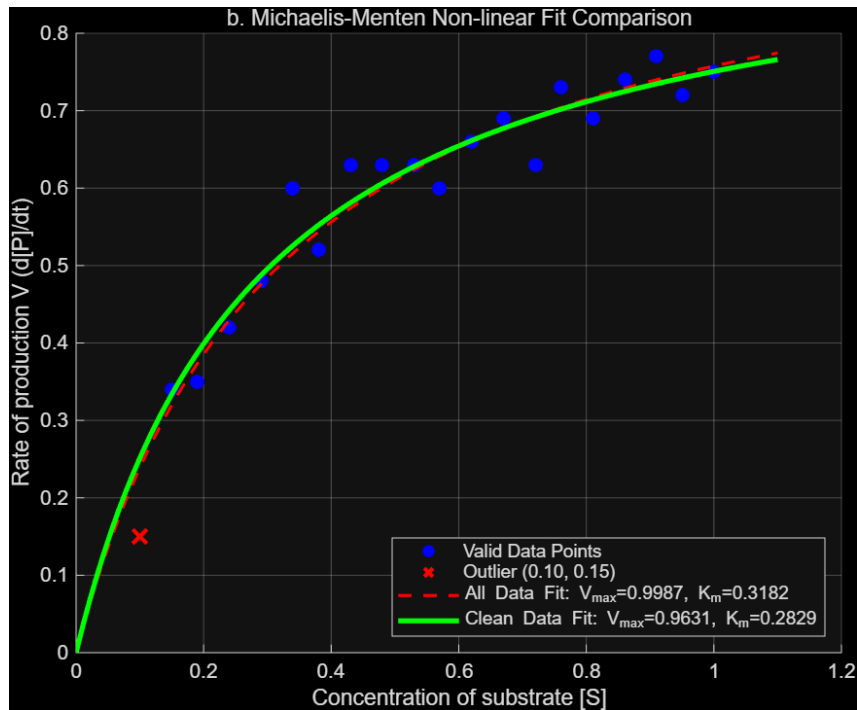


d. yes

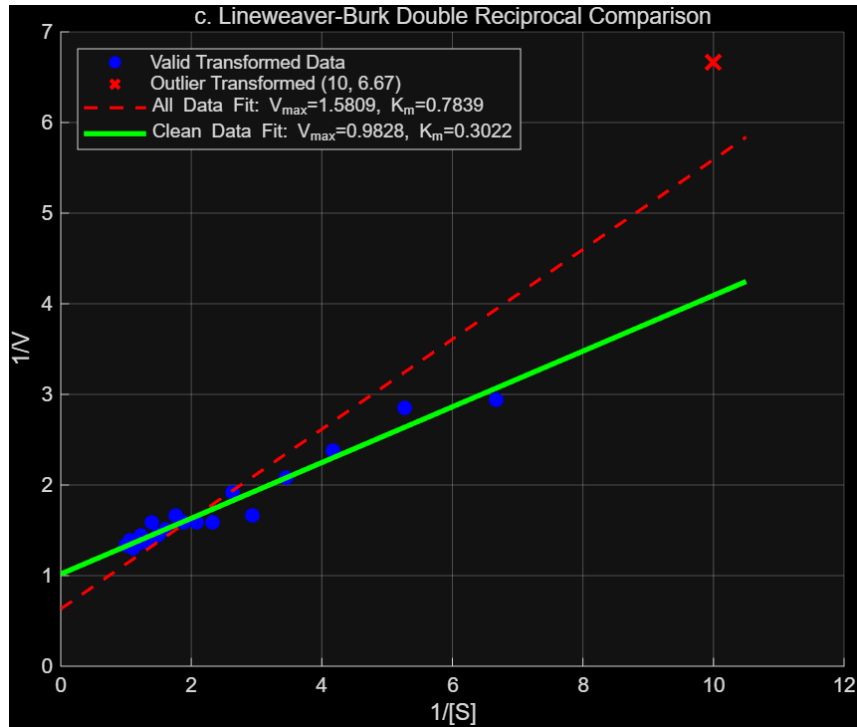
e. 不符合米氏方程的准稳态假设

4.3 a. $V_{max} = k_2[E]_0, K_m = \frac{k_{-1} + k_2}{k_{+1}} = 0.101$

b. $V_{max} = 0.9987, K_m = 0.3182$



c. $V_{max} = 1.5809, K_m = 0.7839$



d. The difference is significant. This is because point (0.15,0.10) is an outlier after taking reciprocal.取倒数后成为离群值 If we rule out this point and fit again, we get $V_{max} = 0.9631$, $K_m = 0.2829$ and $V_{max} = 0.9828$, $K_m = 0.3022$. This greatly makes up for the gap above.